

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 3:

Economic Basis for trade

► What you will learn in this chapter

- *Absolute advantage*
- *Comparative advantage*
- *Gain from trade*

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



Theory of Comparative Advantage:



It makes sense to produce the things you're especially good at producing... and buy everything else from others.

ORIGIN OF THE IDEA

David Ricardo (1772–1823)

Argued against Corn Laws in British Parliament.

Corn Laws: high tariffs on imported grain that protected British landowners.

His *theory of comparative advantage* is still the ideological foundation for free trade.



ORIGIN OF THE IDEA

Donald J Trump

To me, the most beautiful word in the dictionary is tariff. And it's my favorite word.

October 16, 2024



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

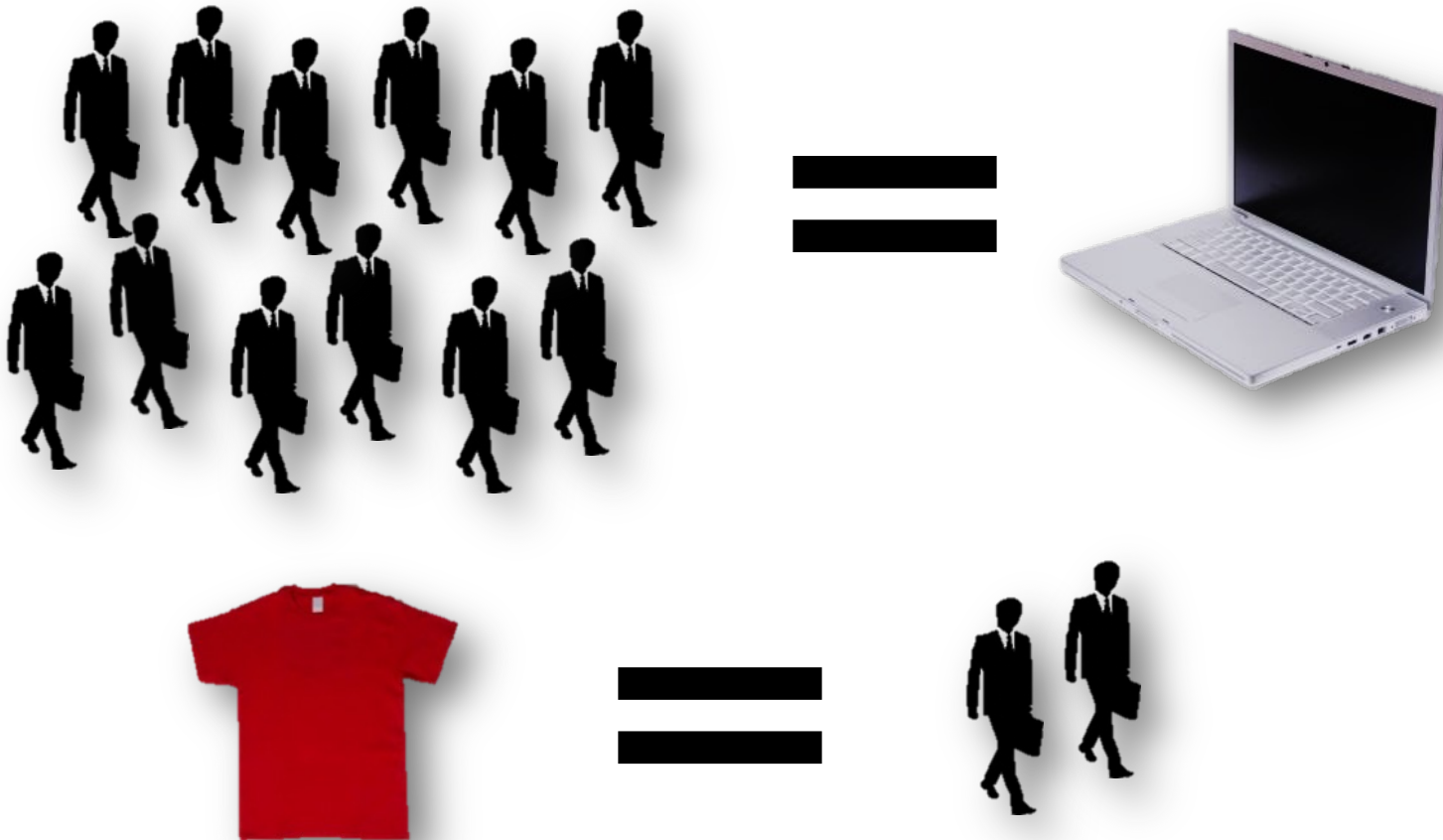
Suppose that there are just two goods, computers and shirts, and one input, labor.



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



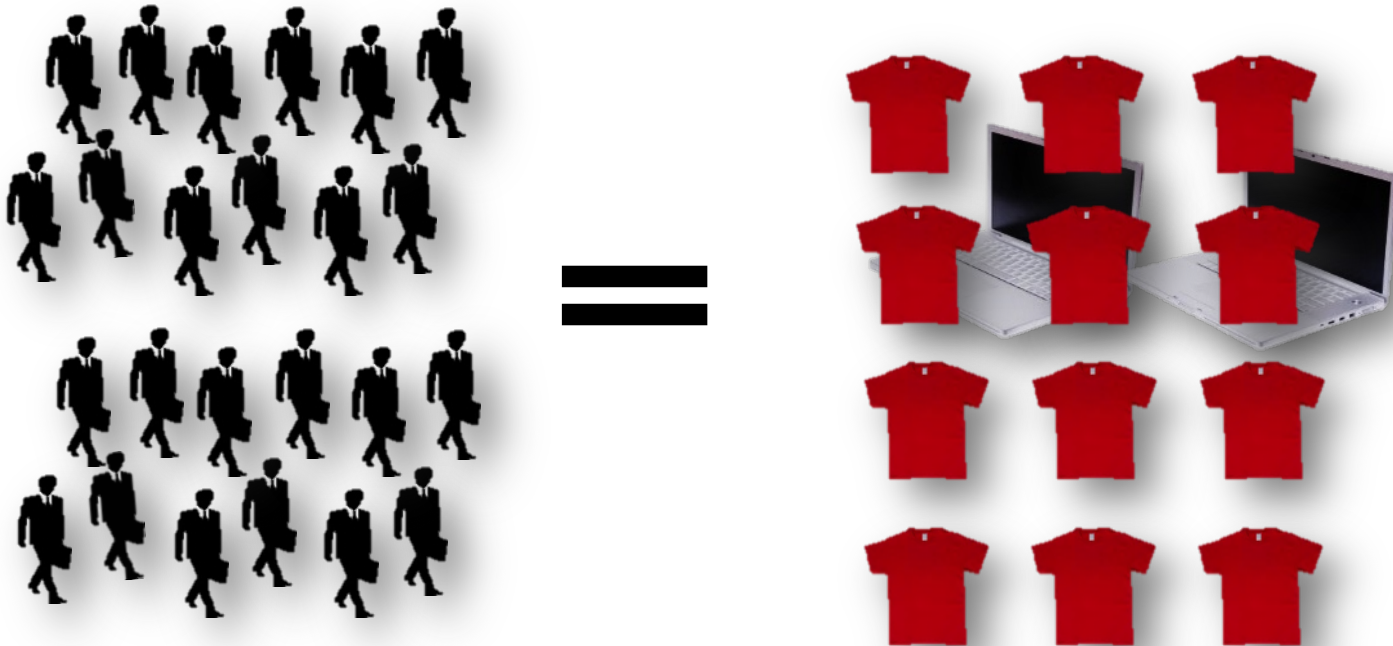
Assume that in Mexico, it takes 12 units of labor to make one computer and 2 units of labor to produce one shirt



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



**Suppose that Mexico has 24 units of labor. It can produce 2 computers and 0 shirts...
or 0 computers and 12 shirts**



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



The PPF is about trade-offs.

If Mexico wants to produce more shirts, it must produce fewer computers.

**Each computer “costs” 12 workers,
who could produce 6 shirts.**

So a computer “costs” 6 shirts in Mexico.



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



AND...

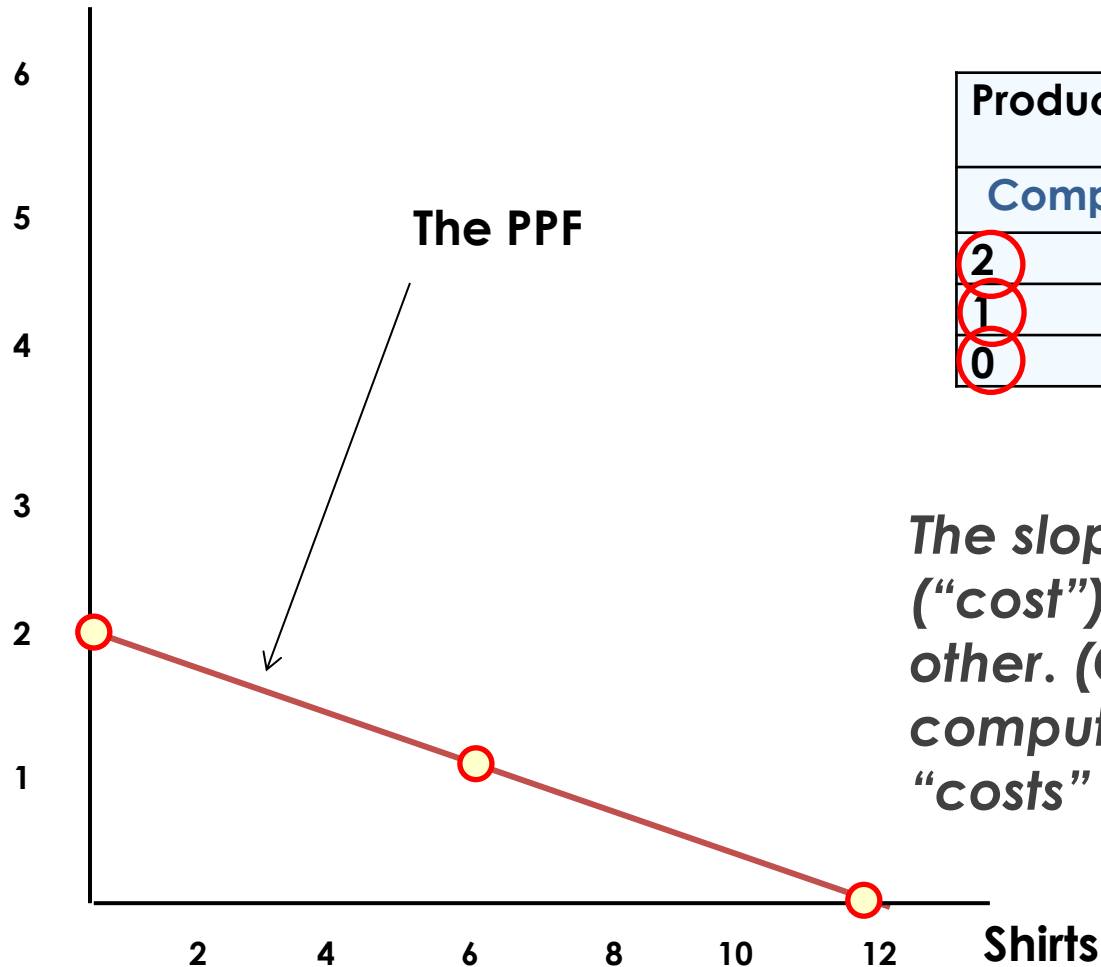
**Each shirt “costs” 2 workers,
who could produce one-sixth of a computer.
So a shirt “costs” one-sixth of a computer in
Mexico.**



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



Computers Mexico's Production Possibilities Frontier



Production Possibilities	
Computers	Shirts
2	0
1	6
0	12

The slope = the trade-off ("cost") of one good for the other. (One-sixth of a computer = 1 shirt: 1 shirt "costs" 6 computers.)

LEARN BY DOING: PRACTICE QUESTION

**Erin has a choice between two activities:
She can repair one transmission per hour or
she can repair two fuel injectors per hour.
What is the opportunity cost of repairing one
fuel injector?**

- a) 1 transmission**
- b) 0.5 transmission**
- c) 2 transmissions**

LEARN BY DOING: PRACTICE QUESTION

David can write either one more book this year or five more articles this year. What is the opportunity cost of writing half of a book this year, in terms of articles?

- a) 0.2 article**
- b) 1 article**
- c) 2.5 articles**
- d) 10 articles**

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

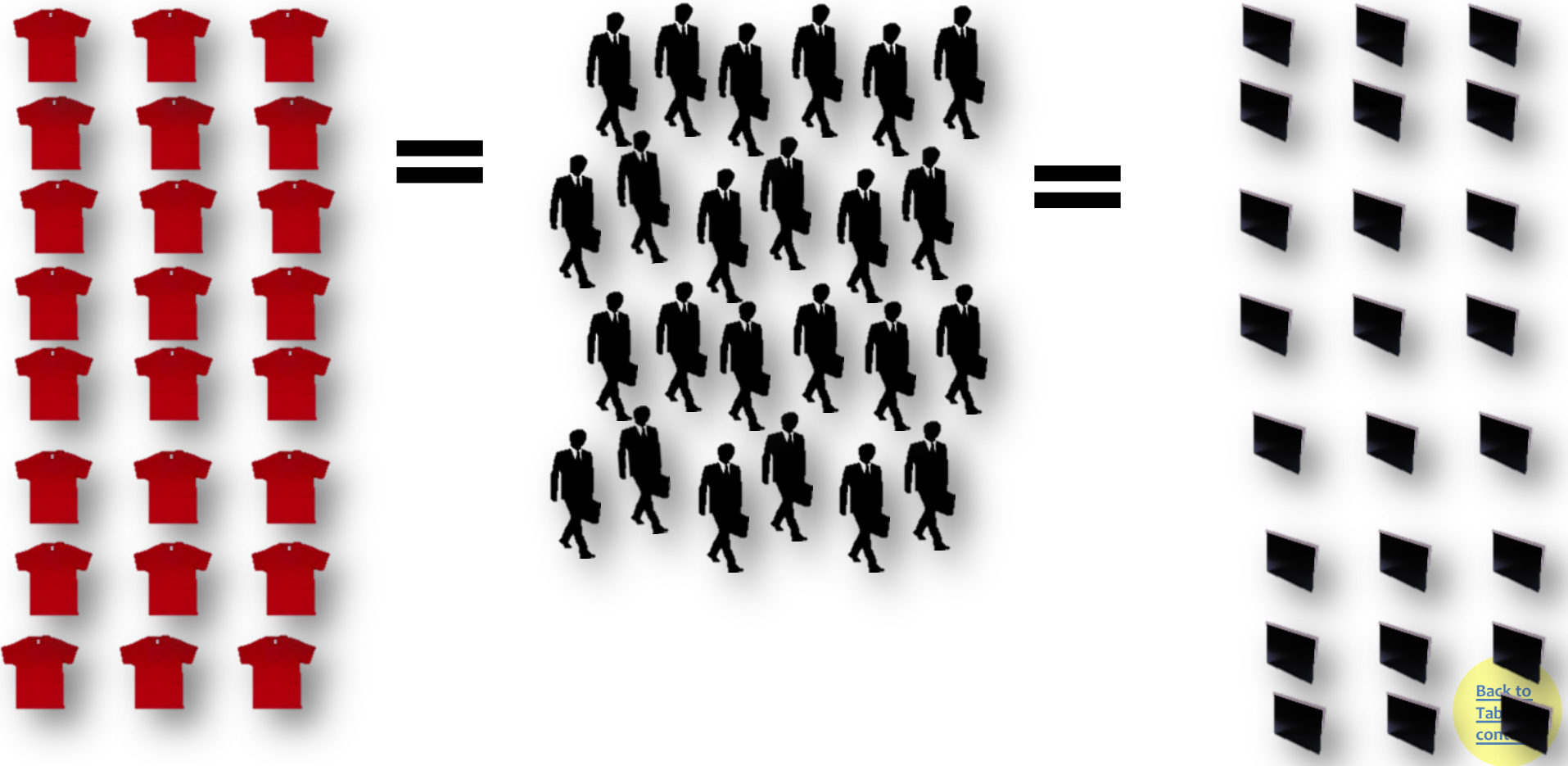
Enter the U.S. with its own production and cost ratios.



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



Assume that in the United States, there are 24 units of labor, and it takes 1 unit of labor to produce either good.



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE



**Each shirt “costs” 1 worker,
who could produce 1 computer.
So a shirt “costs” 1 computer in the United States.**



COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

Summary of costs: each country has a different opportunity cost for each good.

Take a look at the opportunity costs for computers:



**Would there be a benefit to trade?
If so, who should produce what?**

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

A country has a **comparative advantage** in producing goods for which it has the lowest opportunity cost.

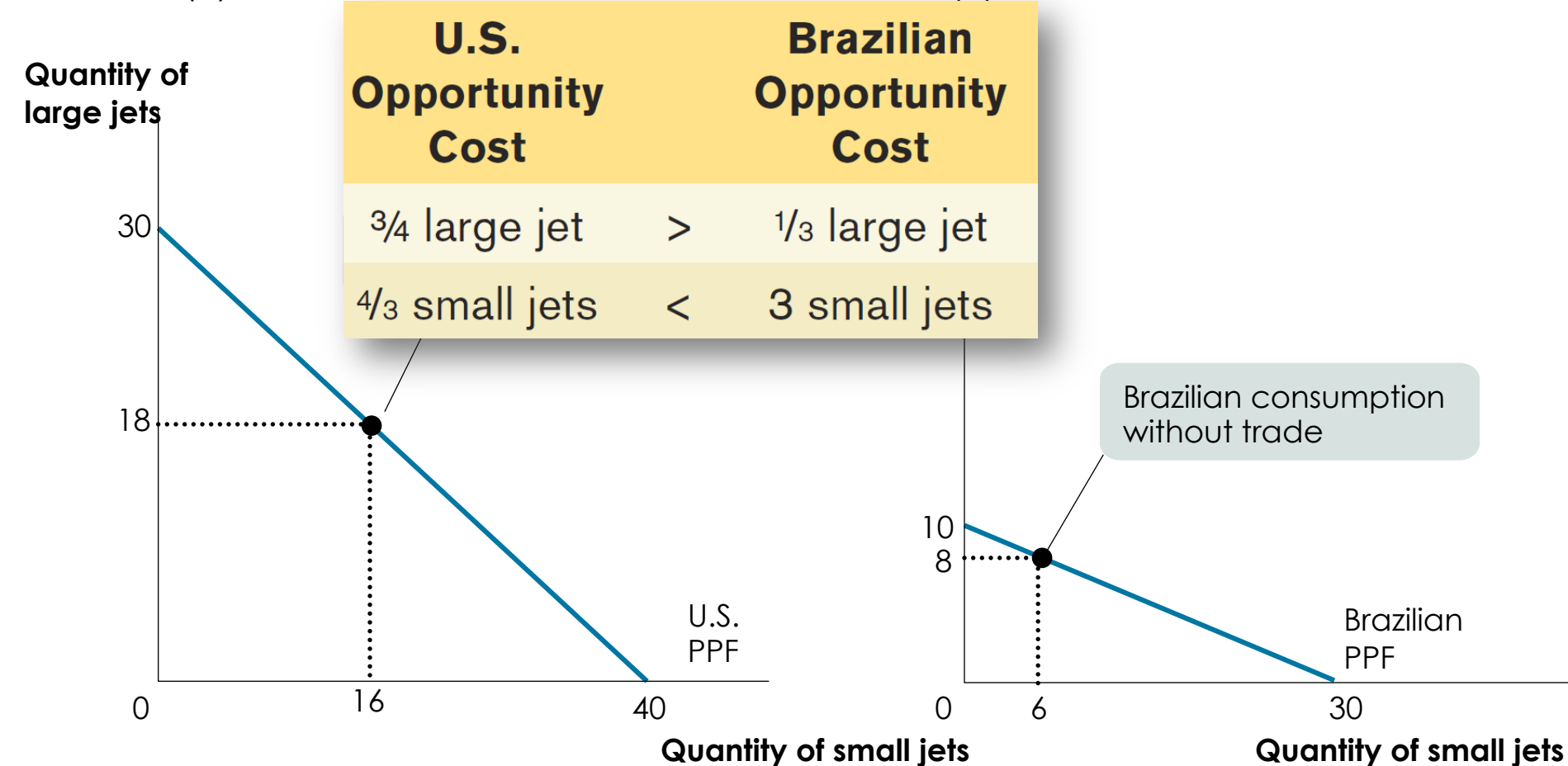
The **U.S. has the comparative advantage in** (is the lower-cost producer of) **computers**: Each computer costs 1 shirt. (Mexico = 6 shirts.)

Mexico has the comparative advantage in (is the lower-cost producer of) **shirts**: Each shirt costs one-sixth of a computer. (United States = 1 computer.)

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

(a) U.S. Production Possibilities

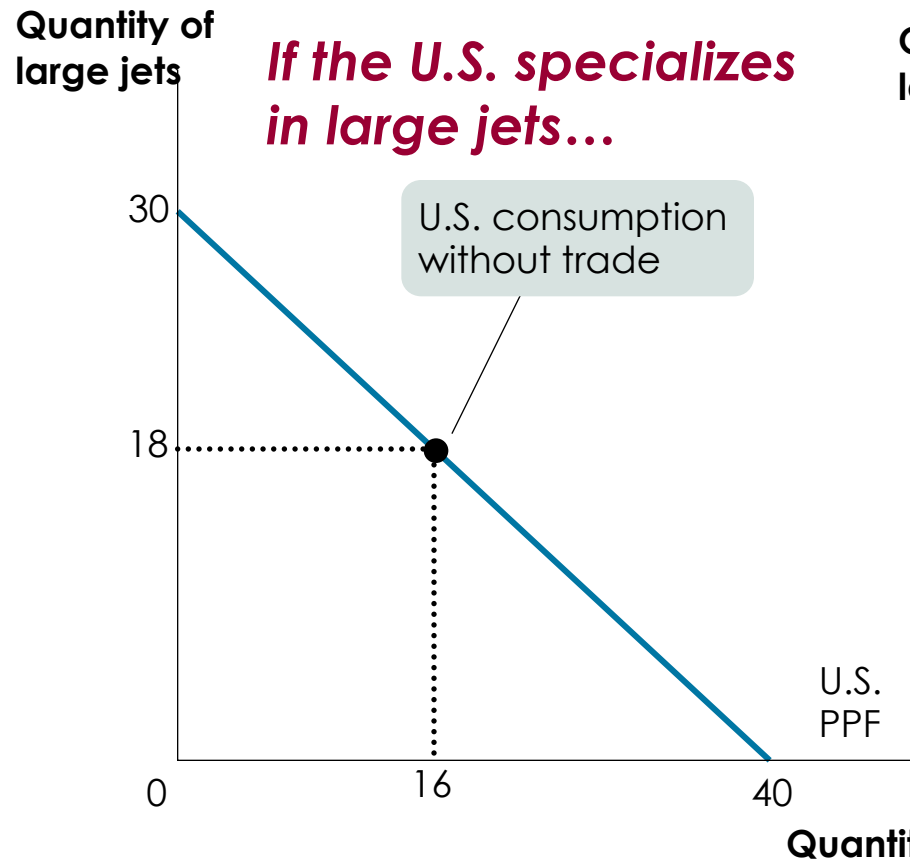
(b) Brazilian Production Possibilities



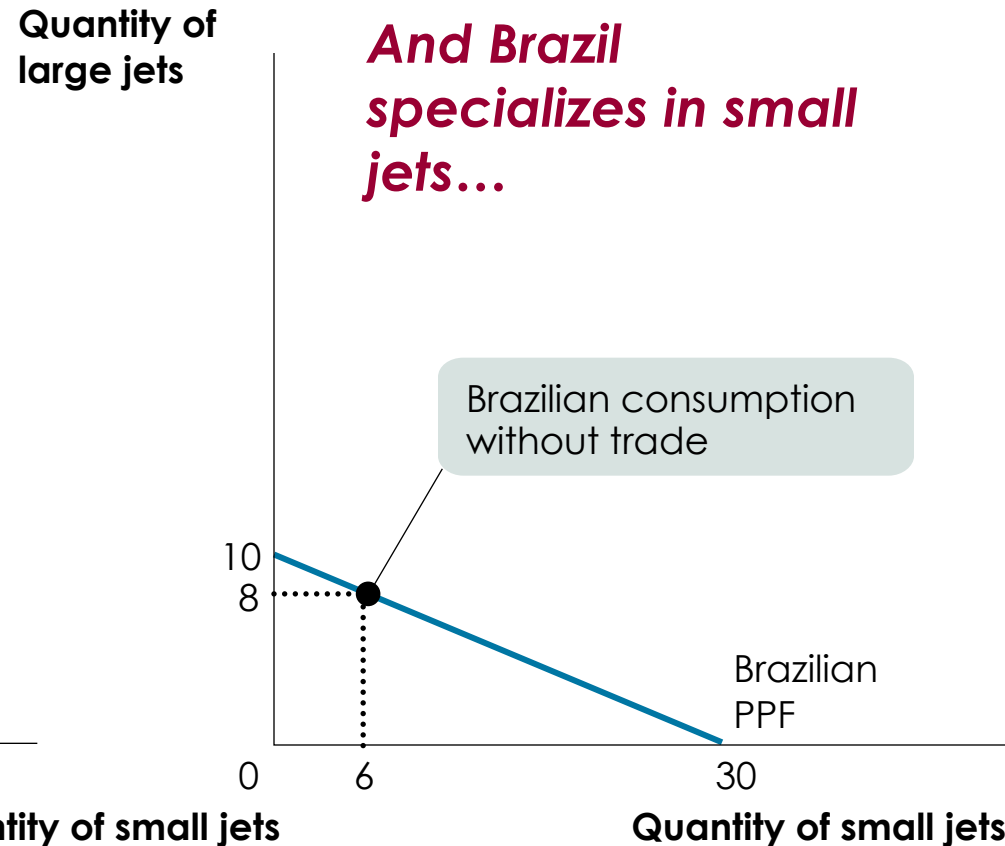
Since each country has a different opportunity cost, it makes sense to specialize and trade...

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

(a) U.S. Production Possibilities



(b) Brazilian Production Possibilities

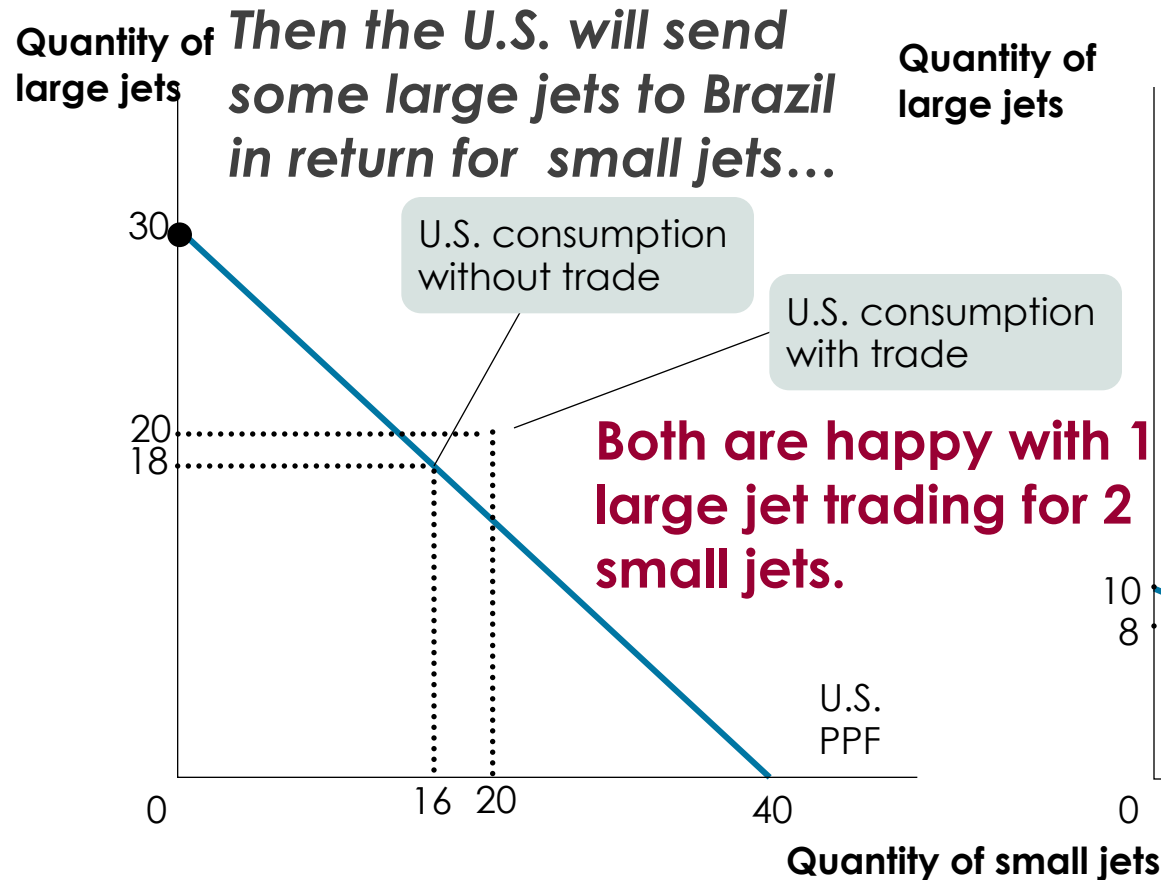


More airplanes will be produced with the same resources (up from 26 ($18 + 8$) to 30 large jets and up from 22 ($16 + 6$) to 30 small jets).

COMPARATIVE ADVANTAGE AND GAINS FROM TRADE

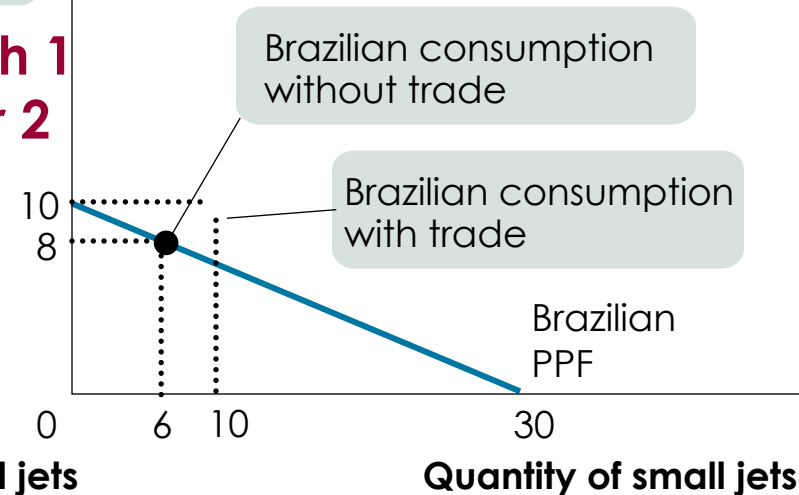
Both countries gain.

(a) U.S. Production Possibilities



(b) Brazilian Production Possibilities

And Brazil will send some small jets to the U.S. in return for large jets...

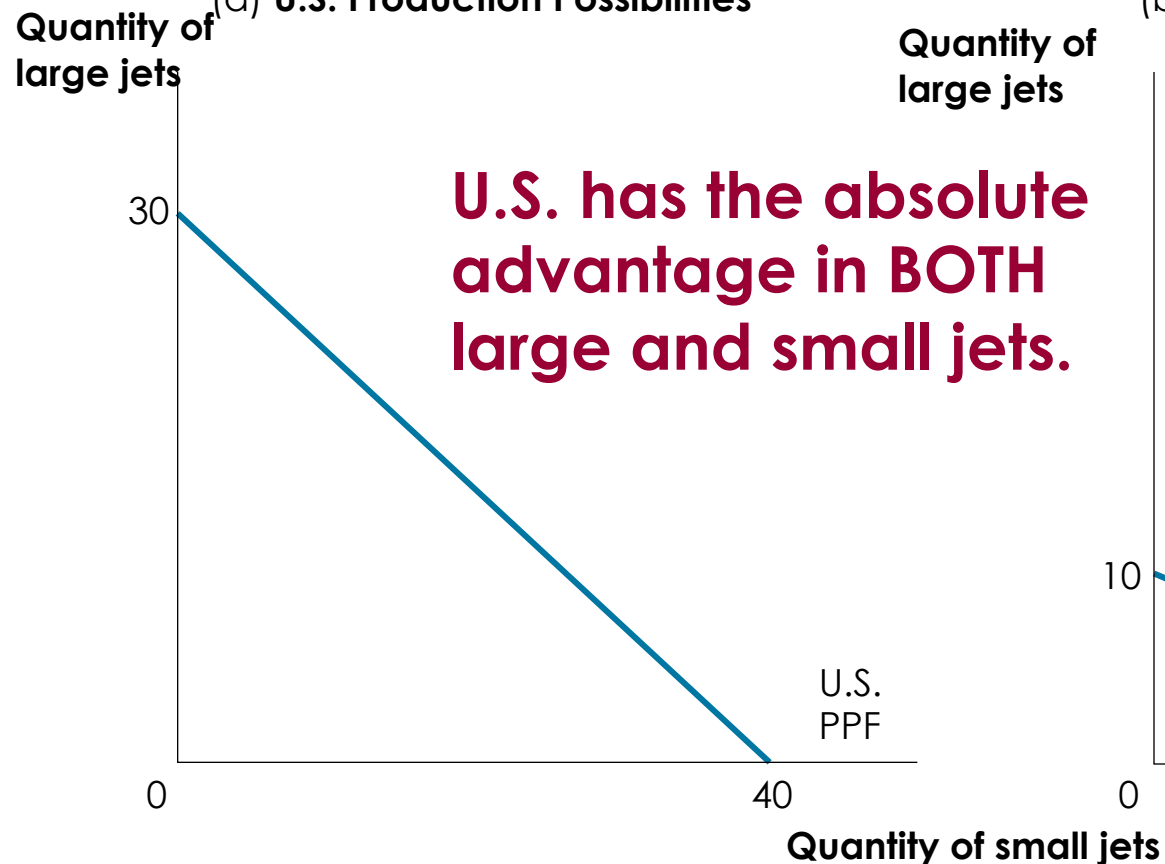


Both countries will be happy to export their goods for any price ABOVE their cost of production and import for any price BELOW their cost of production.

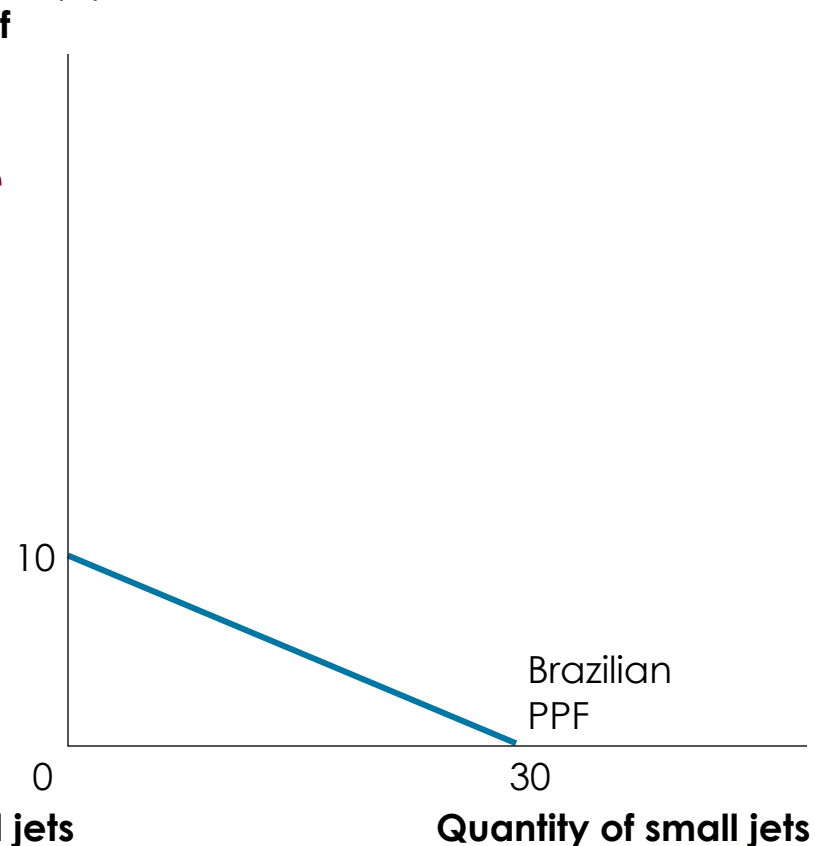
COMPARATIVE ADVANTAGE VS. ABSOLUTE ADVANTAGE

Absolute advantage: a country can produce more output per worker than other countries.

(a) U.S. Production Possibilities



(b) Brazilian Production Possibilities



Don't confuse absolute with comparative advantage.

Just because the United States can produce **more** of both goods doesn't mean we're better off without trade.

Pay attention to **opportunity costs**:

If it's **cheaper** for Brazil to produce small jets than it is for the United States, the United States will want to import small jets from Brazil.



LEARN BY DOING: PRACTICE QUESTION

If **Texia** specializes in food, it can produce **1,000 units of food and 0 units of clothing this year**. If it specializes in clothing, it can produce **500 units of clothing and 0 units of food**. **This year Urbania** can produce either **500 units of food and 0 units of clothing** or **200 units of clothing and 0 units of food**.
(assume linear production possibility frontiers)

_____ has the **absolute** advantage in the production of clothing and _____ has the **absolute** advantage in the production of food.

- a) Texia; Texia
- b) Texia; Urbania
- c) Urbania; Texia
- d) Urbania; Urbania

LEARN BY DOING: PRACTICE QUESTION

If **Texia** specializes in food, it can produce **1,000 units of food and 0 units of clothing this year**. If it specializes in clothing, it can produce **500 units of clothing and 0 units of food**. **This year Urbania** can produce either **500 units of food and 0 units of clothing** or **200 units of clothing and 0 units of food**.
(assume linear production possibility frontiers)

_____ has the **comparative advantage** in the production of clothing and _____ has the **comparative advantage** in the production of food.

- a) Texia; Texia
- b) Texia; Urbania
- c) Urbania; Texia
- d) Urbania; Urbania

LEARN BY DOING: PRACTICE QUESTION

If a country specializes according to its own comparative advantage and then trades with other nations:

- a) it will operate at a point inside its production possibilities frontier.**
- b) it can consume at a higher level than the domestic production possibilities frontier.**
- c) its production possibilities frontier will shift or rotate inward.**
- d) it can consume at the same level as the domestic production possibilities frontier.**

LEARN BY DOING: PRACTICE QUESTION

Write down, then discuss with a partner:

What do you think? Would the U.S. economy be better off if we did not import so much clothing from Bangladesh?

List the groups who you think would gain and the groups who would lose if Bangladesh clothing imports were restricted.

LEARN BY DOING: PRACTICE QUESTION

Factor	Benefit of Restrictions	Cost of Restrictions
Jobs	Potential U.S. manufacturing jobs	Higher consumer prices, job losses in retail/logistics
Prices	—	Inflation in apparel (10–30% price hikes estimated)
Global Equity	—	Harm to Bangladeshi workers (avg. wage: \$113/month)
Supply Chain Resilience	Reduced reliance on distant suppliers	Increased vulnerability to domestic production shocks

LEARN BY DOING: PRACTICE QUESTION

Suppose that in Australia, it takes 2 hours to harvest 10 bushels of apples and 4 hours to harvest 10 bushels of tomatoes. Suppose a worker in Brazil can harvest 10 bushels of apples in 4 hours or 10 bushels of tomatoes in 5 hours. Which of these statements is *TRUE*?

- a) Brazil has a comparative advantage in producing tomatoes.
- b) Brazil has a comparative advantage in producing apples.
- c) Brazil has an absolute advantage in producing both goods.
- d) In trade between these two countries, Australia would gain and Brazil would lose.

LEARN BY DOING: PRACTICE QUESTION

Take a look at how NAFTA works by clicking [here](#) or on the picture below. (5:54 minutes)

